

DETROIT CITY, MI, USA

WMO: 725375

Lat: 42.409N

Lon: 83.010W

Elev: 191

StdP: 99.05

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14822

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-12.9	-21.2	0.6	-14.6	-18.6	0.7	-12.0	10.9	-2.2	9.7	-1.1	4.1	250	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	32.5	22.9	31.0	22.2	29.6	21.4	24.6	29.8	23.6	28.6	22.8	27.6	5.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.1	27.3	22.1	17.1	26.6	21.1	16.2	25.9	75.1	30.0	71.3	28.7	68.1	27.5	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.3	7.6	-18.6	35.1	3.6	1.8	-21.2	36.4	-23.3	37.4	-25.3	38.4	-27.9	39.7		
(5)			-19.2	26.4	3.5	1.1	-21.7	27.2	-23.7	27.9	-25.7	28.5	-28.2	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-3.2	-2.0	2.7	9.3	15.5	21.2	23.6	22.7	18.8	12.1	5.4	-0.1		
	DBStd	10.57	5.95	5.62	5.82	4.91	4.68	3.85	3.00	2.86	4.00	4.59	4.85	5.11		
	HDD10.0	1555	411	336	237	70	6	0	0	0	1	29	152	314		
	HDD18.3	3327	668	568	484	274	114	15	1	3	42	199	387	570		
	CDD10.0	1770	1	1	12	48	178	337	422	394	265	95	15	2		
	CDD18.3	501	0	0	1	2	27	103	165	138	57	7	0	0		
	CDH23.3	4118	0	0	6	31	258	890	1452	1037	403	42	0	0		
	CDH26.7	1269	0	0	1	4	71	304	491	287	106	6	0	0		
(14) Wind	WSAvg	3.7	4.4	4.2	4.1	4.3	3.7	3.4	3.2	3.0	3.1	3.6	4.0	4.1		
(15) Precipitation	PrecAvg	861	60	55	62	76	94	85	81	70	77	69	66	64		
	PrecMax	1123	104	131	106	121	199	143	158	149	155	169	112	113		
	PrecMin	696	21	14	21	15	39	40	33	12	21	11	15	21		
	PrecStd	108	25	30	25	26	40	30	33	36	43	35	23	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	16.0	23.0	27.0	31.1	33.8	34.8	33.3	32.1	27.3	19.9	15.2		
		MCWB	11.6	12.1	15.7	17.2	20.7	23.1	24.6	23.8	22.7	20.0	14.1	12.8		
	2%	DB	10.9	11.5	18.4	23.4	28.7	32.1	32.6	31.2	29.7	24.1	17.0	12.2		
		MCWB	9.1	8.6	12.7	14.9	19.6	22.3	23.1	22.7	21.2	18.0	13.1	10.1		
	5%	DB	7.5	8.0	15.0	20.6	26.4	30.1	31.0	29.7	27.5	21.9	15.0	9.1		
		MCWB	5.4	5.1	10.3	13.4	18.6	21.4	22.4	21.9	20.2	16.8	11.6	6.8		
	10%	DB	4.5	5.2	11.7	17.7	23.8	28.2	29.4	28.1	25.7	19.7	12.8	6.5		
		MCWB	2.8	2.7	7.7	11.6	17.3	20.3	21.5	21.1	19.3	14.9	9.6	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	12.9	16.6	18.6	22.6	24.6	26.2	25.7	24.2	21.2	15.8	13.1		
		MCDB	13.2	14.6	22.6	24.9	28.8	31.7	32.3	31.1	29.7	25.7	18.0	14.6		
	2%	WB	9.1	8.8	13.7	16.4	21.0	23.4	24.7	24.3	22.8	19.5	14.0	10.3		
		MCDB	10.3	10.8	17.1	21.0	26.4	29.9	30.1	28.7	27.3	22.9	16.3	11.7		
	5%	WB	5.3	5.1	11.0	14.3	19.8	22.6	23.8	23.3	21.7	17.4	12.0	7.0		
		MCDB	7.1	7.9	14.4	19.2	24.8	28.4	28.7	27.4	25.6	20.7	14.4	8.9		
	10%	WB	2.8	2.9	7.7	12.3	18.2	21.6	22.8	22.4	20.5	15.7	9.9	4.5		
		MCDB	4.3	5.2	11.3	17.2	23.1	26.6	27.6	26.5	24.2	18.9	12.5	6.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	7.2	8.6	10.1	10.2	9.8	9.6	9.0	9.5	8.6	7.0	5.8		
		MCDBR	9.3	10.6	13.1	14.2	12.6	11.4	10.7	10.1	11.3	11.3	9.8	8.6		
		MCWBR	8.3	8.6	8.8	8.6	6.8	5.4	4.7	4.6	5.2	6.4	7.5	7.3		
	5% WB	MCDBR	9.1	10.4	12.4	12.8	11.3	10.3	9.7	8.9	9.5	9.9	9.2	8.3		
		MCWBR	8.5	8.8	9.3	9.0	7.2	5.5	5.2	4.9	5.1	6.4	7.9	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.334	0.352	0.398	0.429	0.446	0.436	0.428	0.389	0.360	0.320	0.309		
	taud		2.395	2.287	2.353	2.247	2.209	2.201	2.253	2.265	2.355	2.418	2.494	2.420		
	Ebn at Noon		838	858	890	870	848	832	836	831	840	823	815	794		
	Edh at Noon		79	104	110	132	141	142	134	128	109	91	72	72		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.35	3.51	4.62	5.41	6.11	6.10	5.29	4.27	2.77	1.72	1.22		
	RadStd		0.15	0.27	0.35	0.40	0.46	0.38	0.30	0.21	0.37	0.24	0.22	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	N/A	(47)

Nomenclature: See separate page